

Course Name:	Digital Signal and Image Processing	Semester:	VI
Date of Performance:	04 / 03 / 2026	DIV/ Batch No:	A1
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Title: Implement the following Image smoothing & sharpening using spatial domain filters.

- Low pass.
- High pass.
- Median filter.
- High boost filter.

Objective: To learn & understand Image smoothing & sharpening using spatial domain filters.

Expected Outcome of Experiment:

CO	Outcome
CO4	Apply various spatial and frequency domain enhancement techniques for 1-D and 2-D signals

Books/ Journals/ Websites referred:

1. <http://www.mathworks.com/support/>
2. www.math.mtu.edu/~msgocken/intro/intro.html.
3. R. C.Gonsales R.E.Woods, "Digital Image Processing", Second edition, Pearson Education
4. S.Jayaraman, S Esakkirajan, T Veerakumar "Digital Image Processing" Mc Graw Hill.
5. S.Sridhar, "Digital Image processing", oxford university press, 1st edition."

Pre Lab/ Prior Concepts:

Spatial filtering involves passing a weighted mask, or kernel over the image and replacing the original image pixel value corresponding to the centre of the kernel with the sum of the original pixel values in the region corresponding to the kernel multiplied by the kernel weights.

Low pass Filter:

The mean filter replaces each pixel by the average of all the values in the local neighbourhood. The size of the neighbourhood controls the amount of filtering. In a spatial averaging operation, each pixel is replaced by a weighted average of its

neighbourhood pixels. The low-pass filter preserves the smooth region in the image and it removes the sharp variations leading to blurring effect. The 3 by 3 spatial mask which can perform the averaging operation is given below:

3 by 3 low-pass spatial mask = $\frac{1}{9}$ 1 1 1 1 1 1 1 1 1

It is to be noted that the sum of the elements is equal to 1 in the case of a low-pass spatial mask. The blurring effect will be more with the increase in the size of the mask. Normally, the size of the mask will be odd so that the central pixel can be located exactly.

High Pass Filter:

The main aim of image sharpening is to highlight fine details in the image. Image sharpening is used to enhance the high-frequency components. The spatial filter or spatial mask which performs image sharpening is given below: The spatial mask which performs image sharpening is given as

3 by 3 spatial mask = $\frac{1}{9}$ -1 -1 -1 -1 8 -1 -1 -1 -1

Median Filtering:

The **median filter** is a non-linear digital filtering technique, often used to remove noise from an image, signal, and video. Such noise reduction is a typical pre-processing step to improve the results of later processing for example, edge detection on an image. Median filtering is very widely used in digital image processing because, under certain conditions, it preserves edges while removing noise, also having applications in signal processing.

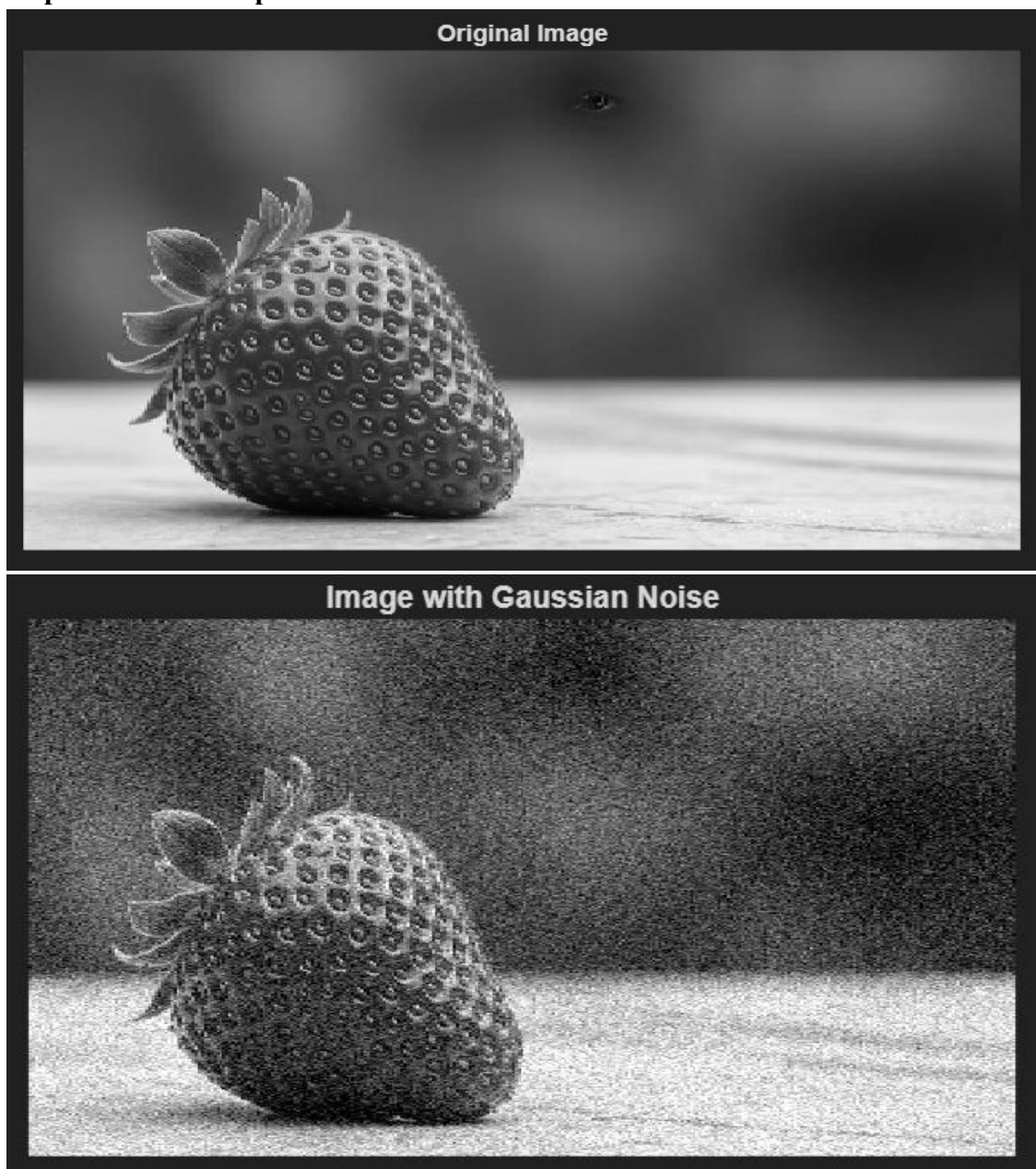
The main idea of the median filter is to run through the signal entry by entry, replacing each entry with the median of the entry and its neighboring entries. The idea is very similar to a moving average filter, which replaces each entry with the arithmetic mean of the entry and its neighbors. The pattern of neighbors is called the "window", which slides, entry by entry, over the entire signal. For one-dimensional signals, the most obvious window is just the first few preceding and following entries, whereas for two-dimensional (or higher-dimensional) data, the window must include all entries within a given radius or ellipsoidal or rectangular region (i.e., the median filter is not a separable filter).

High Boost Filtering:

A high-boost filter is also known as a high-frequency emphasis filter. A high-boost filter is used to retain some of the low-frequency components to aid in the interpretation of an image. In high-boost filtering, the input image $f(m, n)$ is multiplied by an amplification factor A before subtracting the low-pass image. Thus, the high-boost filter expression becomes :

$$\text{High boost} = A \cdot f(m, n) - \text{low pass}$$

Implementation steps with screenshots:



```
img = imread('image.jpg');

if size(img,3) == 3
    img = rgb2gray(img);
end

figure;
imshow(img);
title('Original Image');

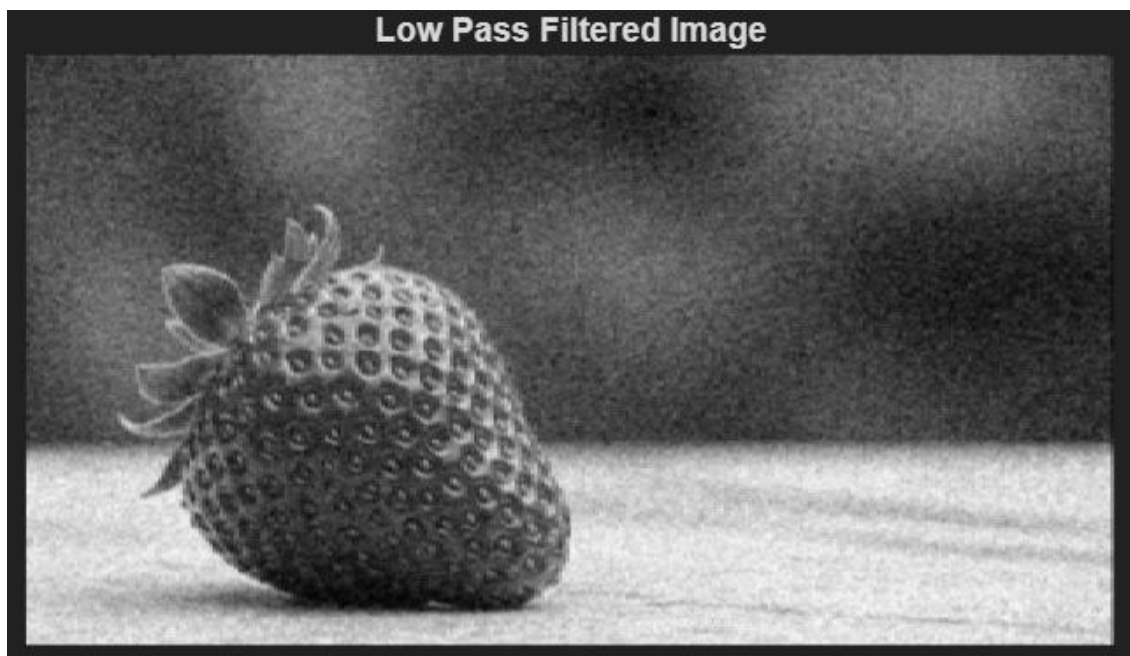
noisy_img = imnoise(img,'gaussian',0,0.01);

figure;
imshow(noisy_img);
title('Image with Gaussian Noise');

mask = (1/9) * [1 1 1;
               1 1 1;
               1 1 1];

filtered_img = imfilter(noisy_img, mask);

figure;
imshow(filtered_img);
title('Low Pass Filtered Image');
```



```
img = imread('image.jpg');  
if size(img,3) == 3  
    img = rgb2gray(img);  
end
```

```
figure;  
subplot(1,2,1);  
imshow(img);  
title('Original Image');  
mask = (1/9) * [-1 -1 -1;  
                -1 8 -1;  
                -1 -1 -1];
```

```
filtered_img = imfilter(img, mask, 'replicate');  
subplot(1,2,2);  
imshow(filtered_img,[]);  
title('High Pass Filtered Image');
```



```
img = imread('image.jpg');  
if size(img,3) == 3  
    img = rgb2gray(img);  
end  
figure;  
subplot(1,2,1);
```

```

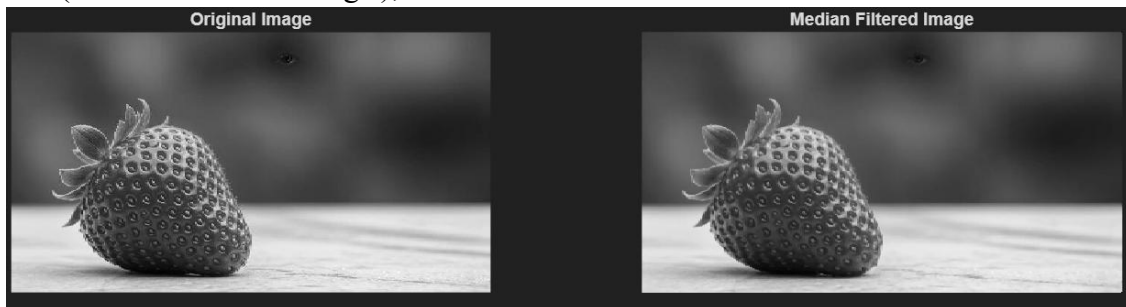
imshow(img);
title('Original Image');

```

```

filtered_img = medfilt2(img,[3 3]);
subplot(1,2,2);
imshow(filtered_img);
title('Median Filtered Image');

```



```

img = imread('image.jpg');
if size(img,3) == 3
    img = rgb2gray(img);
end

```

```

img = double(img);
figure;

```

```

subplot(1,2,1);
imshow(uint8(img));
title('Original Image');

```

```

lowpass = imfilter(img,(1/9)*ones(3,3),'replicate');
A = 1.5;
highboost = A*img - lowpass;
subplot(1,2,2);
imshow(uint8(highboost));
title('High Boost Filtered Image');

```



Conclusion:

I implemented various spatial domain filters for image smoothing and sharpening. Low-pass filtering was used to smooth the image and reduce noise, while high-pass filtering enhanced edges and fine details. Median filtering effectively removed noise while preserving edges. High-boost filtering helped emphasize high-frequency components while retaining important low-frequency information.

Date: 04/03/2026

Signature of faculty in-charge

Post Lab Descriptive Questions

1. What is image smoothing?

Image smoothing is a technique used to reduce noise and small variations in an image. It works by averaging or filtering pixel values in a neighborhood so that sudden intensity changes are reduced. Smoothing produces a blurred image but helps remove unwanted noise.

2. Explain averaging (mean) filter with example.

An averaging or mean filter replaces the value of each pixel with the average of its neighboring pixels. It uses a convolution mask such as:

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Example: If a 3×3 neighborhood contains pixel values:

$$\begin{bmatrix} 10 & 12 & 11 \\ 9 & 10 & 13 \\ 12 & 11 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 12 & 11 \\ 9 & 10 & 13 \\ 12 & 11 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 10 & 12 & 11 \\ 9 & 10 & 13 \\ 12 & 11 & 10 \end{bmatrix}$$

$$\text{Average} = \frac{(10+12+11+9+10+13+12+11+10)}{9} = 10.89 \approx 11$$

The center pixel becomes 11.

3. What is a weighted average filter?

A weighted average filter is similar to the mean filter but assigns different weights to neighboring pixels. Usually, the center pixel has a larger weight. Example mask:

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

This gives more importance to the center pixel and produces smoother results.

4. Explain the median filter and its advantages.

A median filter replaces each pixel with the median value of its neighborhood rather than the average.

Example:

Pixel values in a window: 5, 8, 7, 10, 50, 6, 9, 8, 7

Sorted values: 5, 6, 7, 7, 8, 8, 9, 10, 50

Median = 8

Advantages:

- Removes salt-and-pepper noise effectively
- Preserves edges better than mean filters

- Reduces noise without excessive blurring

5. Compare mean filter and median filter.

Feature	Mean Filter	Median Filter
Type	Linear filter	Non-linear filter
Operation	Uses average of neighbors	Uses median of neighbors
Noise removal	Good for Gaussian noise	Best for salt-and-pepper noise
Edge preservation	Blurs edges	Preserves edges better
Computation	Simple	Slightly more complex

6. Why are median filters effective for salt-and-pepper noise?

Salt-and-pepper noise produces very high or very low intensity values in random pixels. The median filter removes these extreme values by selecting the middle value after sorting the neighborhood pixels, which eliminates outliers while preserving the surrounding pixel information.

7. Solve one example on LPF, HPF and median Filter

Low Pass Filter Example

Original pixels:

10 12 11

9 10 13

12 11 10

Average = $10.89 \approx 11$

Center pixel becomes 11.

High Pass Filter Example

Using mask:

$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$

This enhances edges by emphasizing intensity differences between the center pixel and neighbors.

Median Filter Example

Neighborhood values: 8, 7, 9, 100, 6, 7, 8, 9, 7

Sorted: 6, 7, 7, 7, 8, 8, 9, 9, 100

Median = 8

The noisy value 100 is removed.